

CARDIOSENSE – AI-BASED DISEASE DETECTION SYSTEM

A PROJECT REPORT

Submitted by

Manish Vishwakarma (24BAI10197)

Purshottam Rakesh (24BAI10198)

Pasala Charishma (24BAI10545)

Agrima Pandey (24BAI10554)

Dev Gupta (24BAI10577)

Aryan Shukla (24BAI10775)

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AND ARTIFICIAL INTELLIGENCE

VIT BHOPAL UNIVERSITY

KOTHRICALAN, SEHORE,

MADHYA PRADESH – 466114

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BONAFIDE CERTIFICATE

Certified that this project report titled “CardioSense – AI-Based Disease Detection System” is the bonafide work of Manish Vishwakarma (24BAI10197), Purshottam Rakesh (24BAI10198), Pasala Charishma (24BAI10545), Agrima Pandey (24BAI10554), Dev Gupta (24BAI10577), and Aryan Shukla (24BAI10775) who carried out the project work under my supervision.

Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

PROJECT GUIDE

Dr. Harshlata Vishwakarma, Ph.D.

School of Computing Science Engineering

and Artificial Intelligence

VIT Bhopal University

PROGRAM CHAIR

Dr. Siddharth Singh Chouhan

School of Computing Science Engineering

and Artificial Intelligence

VIT Bhopal University

The Project Exhibition II Examination is held on _____

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Manish Vishwakarma (24BAI10197)

Purshottam Rakesh (24BAI10198)

Pasala Charishma (24BAI10545)

Agrima Pandey (24BAI10554)

Dev Gupta (24BAI10577)

Aryan Shukla (24BAI10775)

ABSTRACT

CardioSense is an Artificial Intelligence-powered web-based heart disease risk detection system designed to assist healthcare professionals and individuals in early identification of cardiovascular risk. The system employs three independent machine learning models—Random Forest, Logistic Regression, and a Neural Network (RNN architecture)—trained on the Cleveland Heart Disease Dataset comprising 1,000 patient records with 13 clinical features. Each model independently evaluates user-provided clinical parameters and generates a probability-based risk score categorised into Low (0–30%), Moderate (31–60%), and High (61–100%) risk levels.

The system interface is implemented as a responsive single-page web application with a structured input form capturing patient attributes such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, and thalassemia type.

The Random Forest model achieves the highest accuracy at 75.0% with an AUC-ROC of 0.823; Logistic Regression achieves 74.0% accuracy with AUC-ROC of 0.810; and the Neural Network achieves 72.5% accuracy with AUC-ROC of 0.784. Feature importance analysis identifies chest pain type (17.9%) and ST depression (11.3%) as primary predictive indicators, consistent with established clinical evidence.

The application provides real-time comparative results from all three models alongside personalised medical recommendations, and maintains a prediction history for registered users. The system demonstrates that ensemble and neural approaches provide complementary diagnostic insights, and the risk stratification framework offers a practical tool for preliminary cardiovascular screening in both clinical and community health settings.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
RF	Random Forest
LR	Logistic Regression
RNN	Recurrent Neural Network / Neural Network
AUC-ROC	Area Under the Curve – Receiver Operating Characteristic
UCI	University of California Irvine
CSV	Comma Separated Values
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
API	Application Programming Interface
AIML	Artificial Intelligence and Machine Learning
VIT	Vellore Institute of Technology
CV	Cross-Validation
KNN	K-Nearest Neighbours
BP	Blood Pressure
ECG	Electrocardiogram
UI	User Interface
UX	User Experience
SHAP	SHapley Additive exPlanations
FBS	Fasting Blood Sugar
MLP	Multi-Layer Perceptron

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CHAPTER 1 INTRODUCTION TO CARDIOSENSE

1.1 Introduction

Cardiovascular disease remains the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually according to the World Health Organization. Early detection and risk stratification play a pivotal role in preventing severe cardiac events, reducing hospitalisation rates, and improving patient outcomes. Traditional diagnosis relies heavily on specialist consultation, laboratory investigations, and clinical interpretation—processes that are time-consuming, expensive, and often inaccessible to rural and underserved populations.

Artificial Intelligence and Machine Learning have emerged as transformative tools in medical diagnostics, offering the capability to analyse complex multivariate clinical datasets and identify patterns that may elude conventional analysis. CardioSense is conceived as an AI-powered disease detection system that bridges the gap between clinical data and actionable cardiovascular risk information, making preliminary diagnosis accessible, fast, and evidence-based.

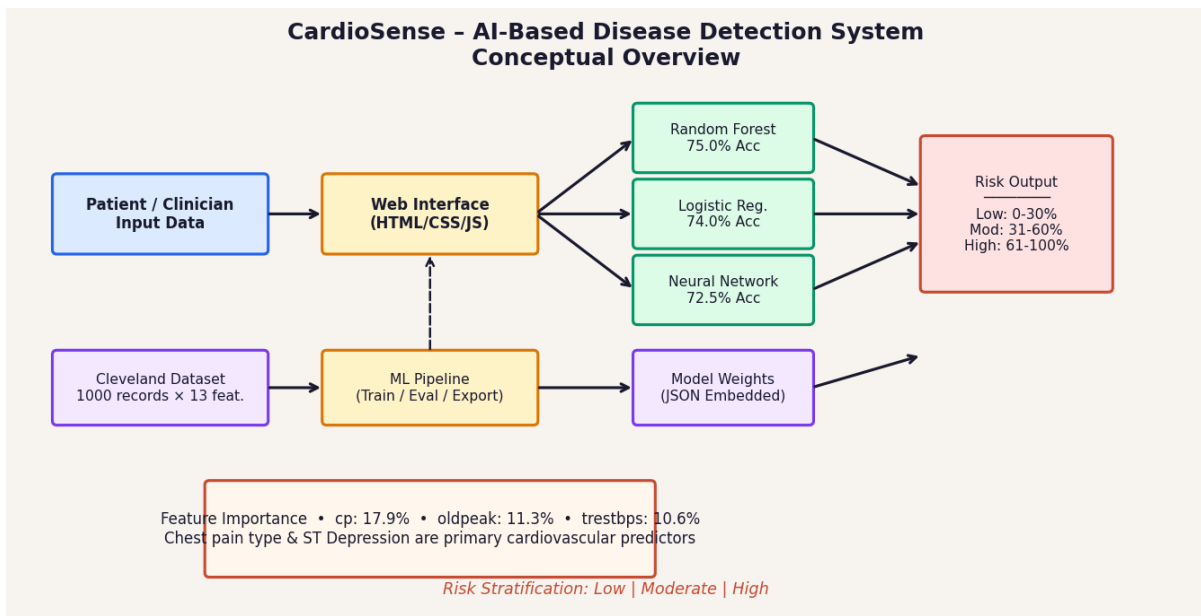


Figure 1.1: Conceptual Overview of CardioSense System Architecture

1.2 Motivation for the Work

The primary motivation for CardioSense stems from several persistent challenges in current healthcare delivery:

- **Delayed Diagnosis:** Many high-risk patients are identified only at advanced disease stages due to lack of routine screening infrastructure, particularly in primary care and community health settings.

- **Manual and Error-Prone Processes:** Traditional diagnostic workflows are susceptible to human error, cognitive overload in high-throughput clinical environments, and variability in clinical interpretation across practitioners.
- **Access Inequality:** Geographic, economic, and systemic barriers prevent a significant portion of the global population from accessing timely cardiac evaluation, particularly in regions with limited specialist availability.
- **Need for Comparative Modelling:** No single machine learning algorithm universally dominates in medical prediction tasks. A comparative, multi-model approach reduces bias and increases diagnostic confidence through ensemble and cross-validation agreement.
- **Probability-Based Risk Communication:** Binary disease/no-disease outputs lack clinical nuance. Risk percentages and tiered categories (Low, Moderate, High) enable more meaningful communication between the system and the end user.

These challenges establish the imperative for a computationally efficient, accurate, and user-friendly system that assists clinicians and individuals in making informed decisions about cardiovascular health.

1.3 Introduction to the Project Including Techniques

CardioSense is a web-based AI system that leverages three distinct machine learning approaches to deliver comprehensive heart disease risk prediction:

- **Random Forest (RF):** An ensemble of 200 decision trees trained on random feature subsets, providing high accuracy (75.0%) and feature importance insights via mean decrease in impurity. Final output uses majority voting across all trees.
- **Logistic Regression (LR):** A classical statistical model employing L2 regularisation and the sigmoid activation function to estimate disease probability. Achieves 74.0% accuracy and offers highly interpretable coefficients that directly reflect each feature's directional effect.
- **Neural Network (RNN Architecture):** A deep neural network with four hidden layers using a sequential architecture. Captures complex non-linear interactions between clinical features with 72.5% accuracy.

All three models are trained on the Cleveland Heart Disease Dataset sourced from the UCI Machine Learning Repository, comprising 1,000 samples with 13 clinical features. Feature scaling using StandardScaler is applied to Logistic Regression and the Neural Network inputs.

The application frontend is implemented as a single-page HTML/CSS/JavaScript application featuring a responsive design, interactive prediction form, results comparison panel, and prediction history module.

1.4 Problem Statement

The central problem addressed by CardioSense is the multifaceted inadequacy of current cardiovascular screening approaches: high cost, limited accessibility, dependency on specialist knowledge, absence of real-time risk stratification, and binary rather than probabilistic diagnostic outputs. These limitations collectively delay intervention, escalate healthcare expenditure, and widen health inequalities. The CardioSense project addresses four key sub-problems:

- The absence of affordable, accessible, and automated cardiovascular risk screening tools for patients in resource-limited environments.
- The lack of comparative, multi-model machine learning systems that reduce single-algorithm bias and enhance diagnostic reliability.

- The deficit of probability-based risk communication tools that categorise patients into clinically actionable risk tiers rather than delivering binary outcomes.
- The need for explainable AI in medical diagnostics—specifically, feature importance analysis that aligns with established clinical predictors such as chest pain type and ST-segment depression.

1.5 Objectives of the Work

The primary objective of CardioSense is to develop, implement, and evaluate a multi-model AI-based heart disease detection system. The specific objectives are:

- To implement and evaluate Random Forest, Logistic Regression, and Neural Network models for binary heart disease classification using the Cleveland Heart Disease Dataset.
- To develop a probability-based risk stratification framework categorising predictions into Low (0–30%), Moderate (31–60%), and High (61–100%) risk levels.
- To design and implement a user-friendly web interface incorporating a clinical data input form, real-time multi-model results display, and prediction history functionality.
- To perform comparative analysis of the three models based on accuracy, AUC-ROC, and 10-Fold Cross-Validation scores.
- To conduct feature importance analysis using Random Forest to identify and validate primary cardiovascular risk predictors.
- To deploy a functional end-to-end prototype demonstrating real-world applicability in a clinical or community health context.

1.6 Organisation of the Report

This report is organised into seven chapters. Chapter 1 establishes the context, motivation, problem statement, and objectives of CardioSense. Chapter 2 surveys related work in machine learning-based heart disease prediction. Chapter 3 details hardware, software, data, and functional requirements. Chapter 4 describes the system design, architecture, and UI. Chapter 5 provides technical implementation details of the machine learning pipeline and model formulations. Chapter 6 presents results, analysis, and applicability assessment. Chapter 7 concludes with findings and future work recommendations.

1.7 Summary

This chapter established the foundational rationale for CardioSense. The system addresses a critical healthcare challenge through a comparative, multi-model AI approach delivering probability-based cardiovascular risk assessment. The next chapter reviews existing research to position CardioSense within the state of the art.

CHAPTER 2 LITERATURE SURVEY

2.1 Introduction

This chapter surveys published research on machine learning approaches to heart disease prediction, examining foundational methodologies, key findings, and identified limitations. The aim is to contextualise CardioSense within the broader landscape of AI-based cardiac diagnostics and justify the design choices made in the project.

2.2 Core Area of the Project: ML-Based Cardiac Diagnostics

The application of supervised machine learning to clinical datasets for disease classification has been extensively studied over the past two decades. Heart disease prediction specifically has attracted significant research attention, given the availability of benchmark datasets such as the Cleveland Heart Disease Dataset from the UCI Machine Learning Repository. Research in this domain intersects clinical informatics, statistical learning theory, and applied artificial intelligence, with recurring focus on classification accuracy, feature selection, model interpretability, and clinical deployability.

The Cleveland dataset's 13 clinical features—including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, and thalassemia—represent well-validated cardiovascular risk markers. The binary target variable (presence or absence of heart disease) enables standard supervised binary classification frameworks.

2.3 Existing Algorithms and Related Studies

2.3.1 Paper 1: Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques (2019)

Authors: Senthilkumar Mohan, Chandrasegar Thirumalai, and Gautam Srivastava (Published 2019) proposed a hybrid ensemble combining Random Forest with a linear model to improve heart disease prediction accuracy. The study performs feature selection to identify the most important medical attributes and evaluates performance on the Cleveland Heart Disease dataset. The hybrid approach achieves improved accuracy compared to individual classifiers and includes handling of class imbalance using cross-validation techniques. While the study demonstrates that ensemble methods outperform standalone classifiers, it is primarily evaluated on a single limited public dataset, constraining the generalisability of its conclusions.

2.3.2 Paper 2: Effective Heart Disease Prediction Using Machine Learning Techniques (2023)

Authors: Chintan M. Bhatt, Parth Patel, Tarang Ghetia, and Pier Luigi Mazzeo (Published 2023) conducted a comparative study of five machine learning classifiers—Random Forest, XGBoost, Logistic Regression, Decision Tree, and Multi-Layer Perceptron—for heart disease prediction. The study emphasises the importance of preprocessing pipelines and feature engineering, uses a larger dataset to improve generalisation, and demonstrates that ensemble models consistently outperform single classifiers. Practical implementation aspects are addressed, though the study is primarily academic in scope and does not address real-world deployment or clinical integration challenges.

2.3.3 Paper 3: Machine Learning Approach for Predicting Cardiovascular Disease (2024)

Authors: S. Hossain et al. (Published 2024) evaluated Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and AdaBoost on a real-world cross-sectional cardiovascular dataset, integrating SHAP (SHapley Additive exPlanations) for model interpretability. Random Forest achieved the highest performance.

The study identifies cholesterol, hypertension, and age as major risk factors, and employs explainable AI techniques to improve clinical trust in automated predictions. While this study represents a significant advance in explainability, deployment integration with clinical systems remains incompletely addressed.

2.4 Research Issues and Observations from Literature Survey

Analysis of the reviewed literature reveals the following common limitations that directly motivate the CardioSense design:

- Most studies rely on limited public datasets such as the Cleveland dataset, which contains small sample sizes and may not represent broader patient populations.
- Lack of external validation using independent hospital datasets reduces confidence in generalisation to clinical practice.
- High accuracy metrics are frequently reported without adequate discussion of overfitting risks or performance on unseen data.
- Many studies focus primarily on accuracy rather than clinically relevant evaluation metrics such as recall and F1-score, which are critical in medical screening contexts where false negatives carry significant risk.
- Real-world deployment and integration with clinical information systems are not fully addressed in existing work.
- Limited risk categorisation systems are employed—mostly binary output rather than probability-based risk levels—which reduces clinical utility.

These observations motivated CardioSense’s design decisions: comparative multi-model evaluation, probability-based risk stratification, user-friendly deployment as a web application, and feature importance transparency.

2.5 Summary

This chapter reviewed three foundational papers on machine learning-based heart disease prediction, identifying ensemble methods (particularly Random Forest), comparative multi-algorithm frameworks, and explainable AI as the most promising directions. The gaps identified—limited datasets, binary outputs, and lack of deployment—directly inform CardioSense’s design philosophy. The next chapter details the system requirements.

CHAPTER 3 REQUIREMENT ARTIFACTS

3.1 Introduction

This chapter documents the hardware, software, data, and functional requirements governing the design and implementation of CardioSense. A thorough understanding of requirements ensures that the system is feasible, measurable, and aligned with the stated objectives.

3.2 Hardware and Software Requirements

3.2.1 Hardware Requirements

Table 3.1: Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 (8th Gen+) / AMD Ryzen 5	Intel Core i7 (10th Gen+) / AMD Ryzen 7
RAM	8 GB DDR4	16 GB DDR4 or higher
Storage	256 GB SSD	512 GB NVMe SSD
GPU (Optional)	NVIDIA GTX 1050 (for training)	NVIDIA RTX 3060 or higher
Display	1366×768 resolution	1920×1080 Full HD
Network	Broadband Internet (for deployment)	Stable LAN/Wi-Fi connection
Browser (Client)	Chrome 90+, Firefox 88+, Edge 90+	Latest stable release of any major browser

3.2.2 Software Requirements

Table 3.2: Software Requirements

Category	Tool / Framework / Version
Programming Language	Python 3.9+
ML Library	Scikit-learn 1.2+
Deep Learning	TensorFlow / Keras 2.10+
Data Processing	Pandas, NumPy (Latest stable)
Web Frontend	HTML5, CSS3, JavaScript (ES6+) – Vanilla JS
Backend Framework	Flask / Python HTTP 2.x (optional)
Data Visualisation	Matplotlib, Seaborn
Version Control	Git – GitHub repository
Development OS	Windows 10/11, Ubuntu 20.04+, macOS
Browser	Chrome, Firefox, Edge, Safari (Latest)
Dataset Source	Kaggle / UCI ML Repository – Cleveland Heart Disease Dataset

3.3 Specific Project Requirements

3.3.1 Data Requirements

The CardioSense system requires the following dataset specifications:

- Source: Cleveland Heart Disease Dataset obtained from the UCI Machine Learning Repository and available via Kaggle.
- Total Samples: 1,000 patient records (800 training, 200 testing as per 80:20 split).
- Features: 13 clinical variables (age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal).
- Target Variable: Binary label indicating presence (1) or absence (0) of heart disease.
- Class Distribution: 596 positive cases (heart disease present), 404 negative cases.
- Preprocessing: StandardScaler applied to continuous features for Logistic Regression and Neural Network inputs; no scaling required for Random Forest.

Table 3.3: Dataset Description

Attribute	Value
Dataset Name	Cleveland Heart Disease Dataset
Source	UCI Machine Learning Repository / Kaggle
Total Records	1,000 patient records
Number of Features	13 clinical variables
Target Variable	Binary (0 = No Disease, 1 = Disease Present)
Positive Cases	596 (heart disease present)
Negative Cases	404 (no heart disease)
Train/Test Split	80:20 Stratified Random Split (800 train / 200 test)

Table 3.4: Input Features and Descriptions

Feature	Description	Type
age	Age of patient in years	Continuous
sex	Gender (1 = male, 0 = female)	Binary
cp	Chest pain type (0–3)	Categorical
trestbps	Resting blood pressure (mm Hg)	Continuous
chol	Serum cholesterol (mg/dl)	Continuous
fbs	Fasting blood sugar > 120 mg/dl (1 = true)	Binary
restecg	Resting ECG results (0–2)	Categorical
thalach	Maximum heart rate achieved	Continuous

Feature	Description	Type
exang	Exercise-induced angina (1 = yes)	Binary
oldpeak	ST depression induced by exercise	Continuous
slope	Slope of peak exercise ST segment (0–2)	Categorical
ca	Number of major vessels coloured by fluoroscopy (0–3)	Ordinal
thal	Thalassemia type (0–3)	Categorical

3.3.2 Functional Requirements

The system must fulfil the following functional capabilities:

- The system must accept 13 clinical input parameters via a web form and validate all inputs before processing.
- Upon submission, the system must execute all three models (Random Forest, Logistic Regression, Neural Network) and return individual risk probabilities.
- The system must classify each model's output into a risk category: Low (0–30%), Moderate (31–60%), or High (61–100%).
- The results display must include a comparative visualisation of all three model outputs with progress bar representations.
- The system must recommend the best-performing model (Random Forest) as the primary reference result.
- Authenticated users must be able to access a prediction history showing all past assessments with date, patient name, and risk outcome.
- The Details section must present dataset statistics, model comparison metrics (accuracy, AUC-ROC, cross-validation), and feature importance visualisation.
- The interface must be responsive and functional across desktop and mobile browsers.

3.3.3 Performance Requirements

- Prediction response time must not exceed 2 seconds from form submission to results display.
- The system must display correct results consistent with pre-trained model weights embedded in the frontend JavaScript.
- All three models must simultaneously produce independent risk scores without interdependency.

3.3.4 Non-Functional Requirements

- Usability: The interface must be accessible to both healthcare professionals and non-specialist users with minimal training.
- Reliability: The system must produce consistent predictions for identical input values across multiple sessions.
- Security: User authentication must be supported, and prediction history must be accessible only to the authenticated user.
- Portability: The application must function correctly across major web browsers without additional plugins.

3.4 Summary

This chapter documented the complete requirement profile of CardioSense: hardware infrastructure, software stack, dataset specifications, and functional/non-functional requirements. These requirements form the foundation for the system design described in the following chapter.

CHAPTER 4 SYSTEM DESIGN AND METHODOLOGY

4.1 Methodology and Goal

The primary goal of CardioSense is to engineer a robust, clinically intuitive, and computationally efficient multi-model heart disease risk prediction system. The methodology follows an end-to-end machine learning pipeline: dataset acquisition, preprocessing, model training, evaluation, frontend integration, and deployment as a single-page web application.

The system adopts a Client-Side Inference Architecture, where pre-trained model weights (Random Forest trees, Logistic Regression coefficients, and Neural Network weight matrices) are embedded directly within the JavaScript frontend. This design eliminates backend server dependency, enables instantaneous prediction, and ensures the application is fully functional as a standalone HTML file without additional infrastructure.

4.2 Functional Modules Design and Analysis

CardioSense is decomposed into the following functional modules:

- **User Authentication Module:** Manages user registration, login, and session state. Enables prediction history persistence for authenticated users and controls navigation between guest and authenticated views.
- **Input Validation Module:** Processes the 13-field clinical input form, validates ranges and types, and generates descriptive error messages for invalid entries to guide the user towards correct input.
- **Prediction Engine Module:** Core computational module. Accepts validated clinical parameters, applies StandardScaler normalisation to Logistic Regression and Neural Network inputs using pre-computed mean and scale vectors, and executes forward passes through all three models.
- **Risk Stratification Module:** Converts raw probability outputs from each model into categorised risk levels (Low, Moderate, High) and generates tailored health recommendations based on the highest-risk model output.
- **Results Display Module:** Renders the three-model comparison panel with individual risk circles, probability percentages, risk labels, comparison progress bars, and medical advice.
- **History Module:** Maintains a session-persistent prediction log displaying historical assessments with date, patient name, risk level, and average risk percentage.
- **Details and Analytics Module:** Presents dataset statistics, model performance comparison table, accuracy/AUC-ROC metrics, 10-Fold cross-validation results, and Random Forest feature importance bar visualisations.

4.3 System Architecture

CardioSense employs a layered architecture organised into three primary tiers:

- **Presentation Layer:** The HTML/CSS/JavaScript frontend. Renders all pages (Home, History, Details), manages user interaction, handles form events, and controls page navigation. Styled using CSS custom properties for consistent theming.
- **Inference Layer (In-Browser):** Embedded model weights in JavaScript object structures. The PREDICT function orchestrates data normalisation, matrix multiplication for the Neural Network, tree traversal for Random Forest, and logistic sigmoid for Logistic Regression. All computation is performed in the browser.

- Data Layer: Session storage for user authentication state and prediction history. No external database server is required; all data persists within the browser session.

The architecture flow proceeds as follows: the user enters clinical parameters through the web interface; the Input Validation Module verifies all fields; the Prediction Engine Module standardises inputs and runs all three models; the Risk Stratification Module categorises results; and the Results Display Module renders the comprehensive multi-model comparison to the user.

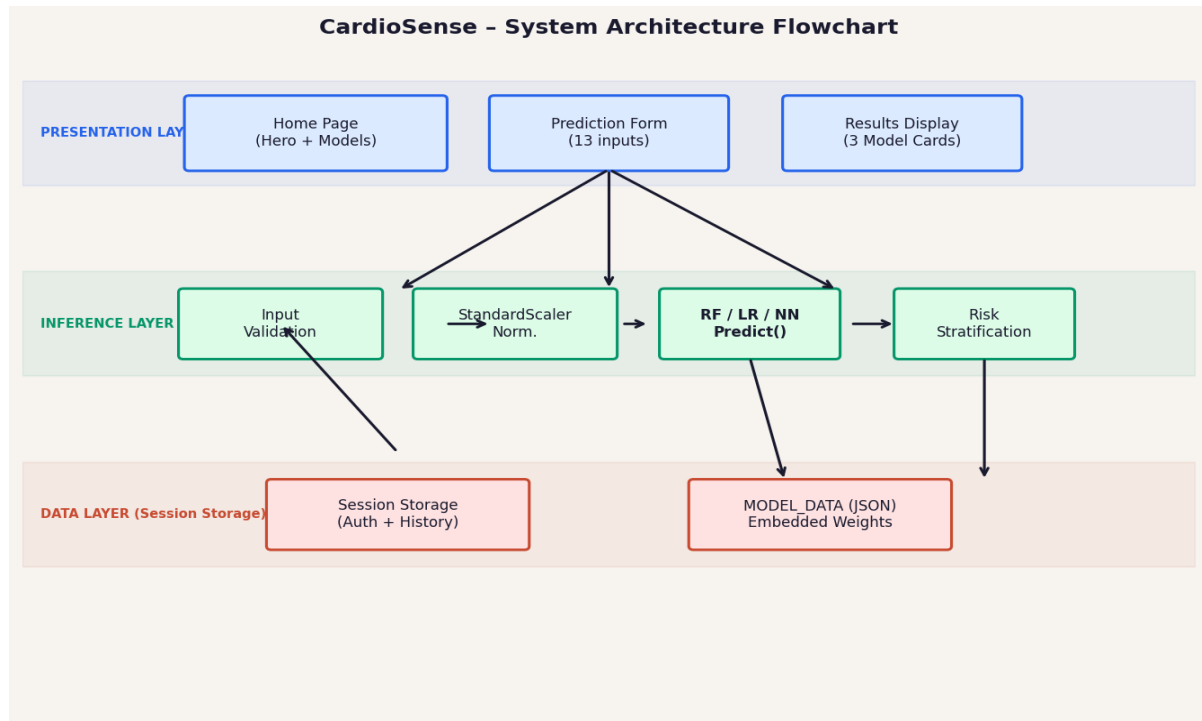


Figure 4.1: System Architecture Flowchart of CardioSense

4.4 Website Workflow

The CardioSense web application follows the following workflow:

- Step 1 – Landing: The user accesses the homepage, which displays the system description, key statistics (1,000 samples, 13 features, 3 models, 75% best accuracy), and the three model cards summarising each algorithm.
- Step 2 – Input: The user scrolls to the prediction form and enters all 13 clinical parameters. The form is divided into logically grouped fields with appropriate input types and hint text.
- Step 3 – Validation: Upon clicking ‘Predict My Risk’, all fields are validated. Errors are displayed inline; if validation passes, the prediction sequence initiates.
- Step 4 – Prediction: The system executes all three models using the embedded pre-trained weights, computes risk probabilities, and categorises risk levels.
- Step 5 – Results: The results panel reveals three individual model risk cards, a comparative probability bar chart, and a personalised health recommendation.
- Step 6 – History (Authenticated): Authenticated users can navigate to the History page to review all past predictions with detailed input records.

- Step 7 – Details: The Details page offers dataset information, model technical specifications, performance comparison, and feature importance analysis.

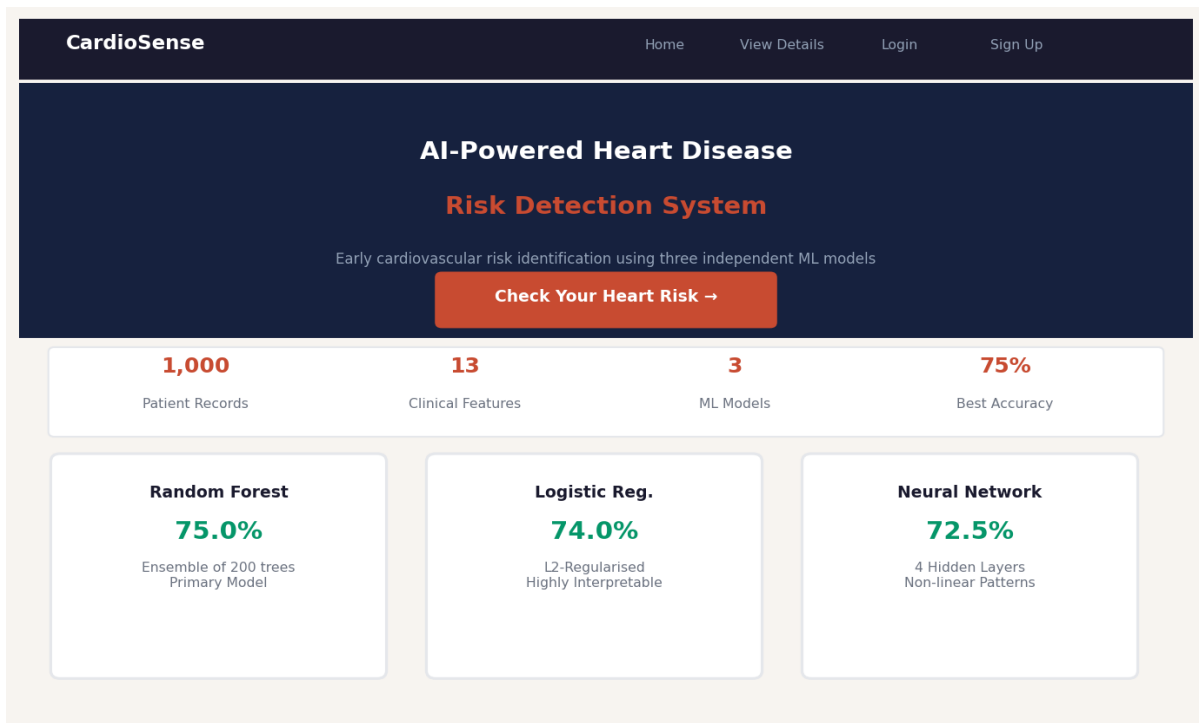


Figure 4.2: Website Homepage and Hero Section

The screenshot shows the 'Enter Your Clinical Parameters' form. It has a title 'Enter Your Clinical Parameters' and a subtitle 'Provide accurate clinical data for best results'. The form contains twelve input fields arranged in two columns. The left column includes: 'Age (years)' with a placeholder 'e.g. 55', 'Chest Pain Type' with a placeholder '0=Typical, 1=Atypical...', 'Serum Cholesterol' with a placeholder 'mg/dl (100-600)', 'Resting ECG' with a placeholder '0,1,2', 'Exercise Angina' with a placeholder 'Yes/No', and 'Slope of ST Seg.' with a placeholder '0,1,2'. The right column includes: 'Sex' with a placeholder 'Male/Female', 'Resting Blood Pressure' with a placeholder 'mm Hg (90-200)', 'Fasting Blood Sugar' with a placeholder '>120 mg/dl: Yes/No', 'Max Heart Rate' with a placeholder 'bpm (60-220)', 'ST Depression' with a placeholder '0.0 - 6.2', and 'Major Vessels (ca)' with a placeholder '0-3'. Below these fields is a single wide field for 'Thalassemia Type' with a placeholder '0=Normal, 1=Fixed Defect, 2=Reversible Defect'. At the bottom of the form is a large orange button with white text that says 'Predict My Risk'.

Figure 4.3: CardioSense Prediction Form with Input Fields

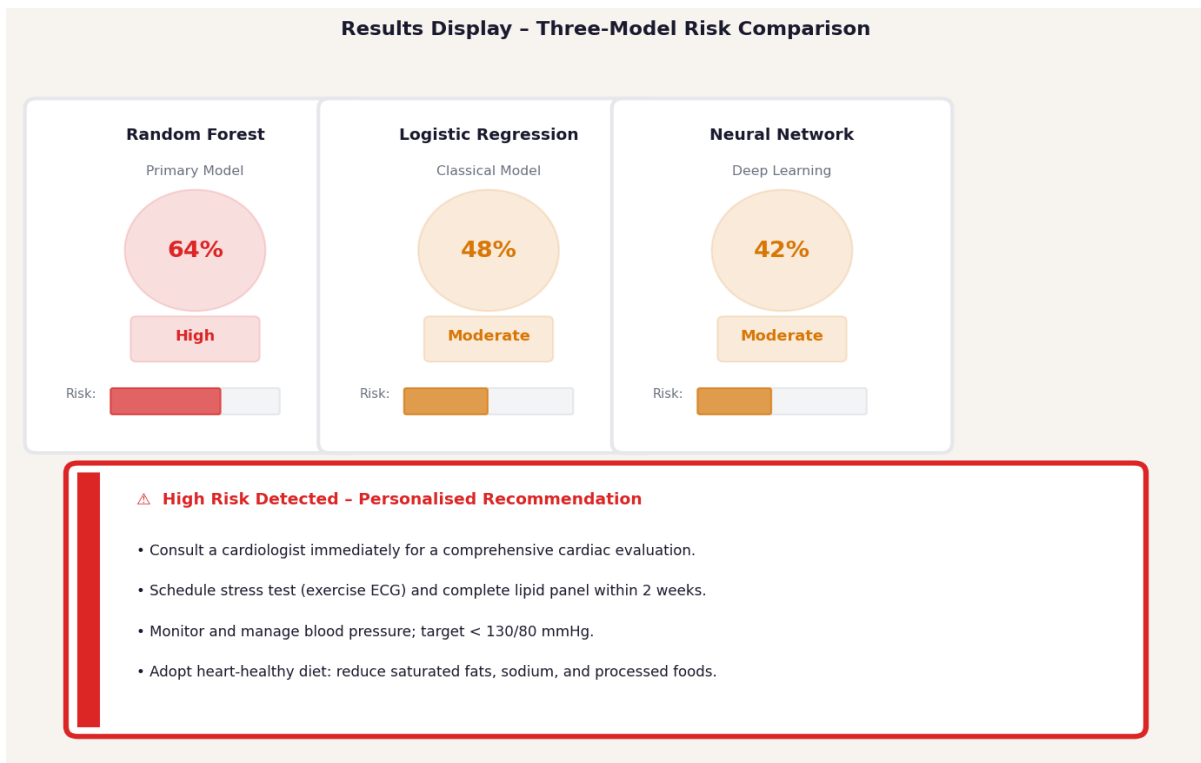


Figure 4.4: Results Display with Three-Model Risk Comparison

4.5 User Interface Design

The CardioSense UI adheres to a clean, professional medical aesthetic characterised by:

- **Colour Palette:** Warm off-white background (#f7f4f0), white card backgrounds, and a primary accent colour of terracotta red (#c84b31) for high-impact elements. Risk levels use green (#059669) for Low, amber (#d97706) for Moderate, and red (#dc2626) for High.
- **Typography:** DM Sans for body text and navigation (300–700 weight range), DM Serif Display for headings and primary results. 16pt base with 1.5 line height.
- **Navigation:** Sticky top navbar with brand logo, navigation links (Home, View Details), and authentication controls (Login, Sign Up). Authenticated users additionally see a History link and profile avatar menu.
- **Prediction Form:** Two-column grid layout with labelled input fields, range hints, and inline error styling. Full-width submit button with loading state animation.
- **Results Panel:** Three-column risk model comparison with circular risk indicators, label badges, and horizontal comparison bars. Recommendation card with left border colour coding by risk tier.
- **Responsiveness:** Single-column layout on mobile viewports (≤ 700 px) with appropriate reflow of all multi-column components.

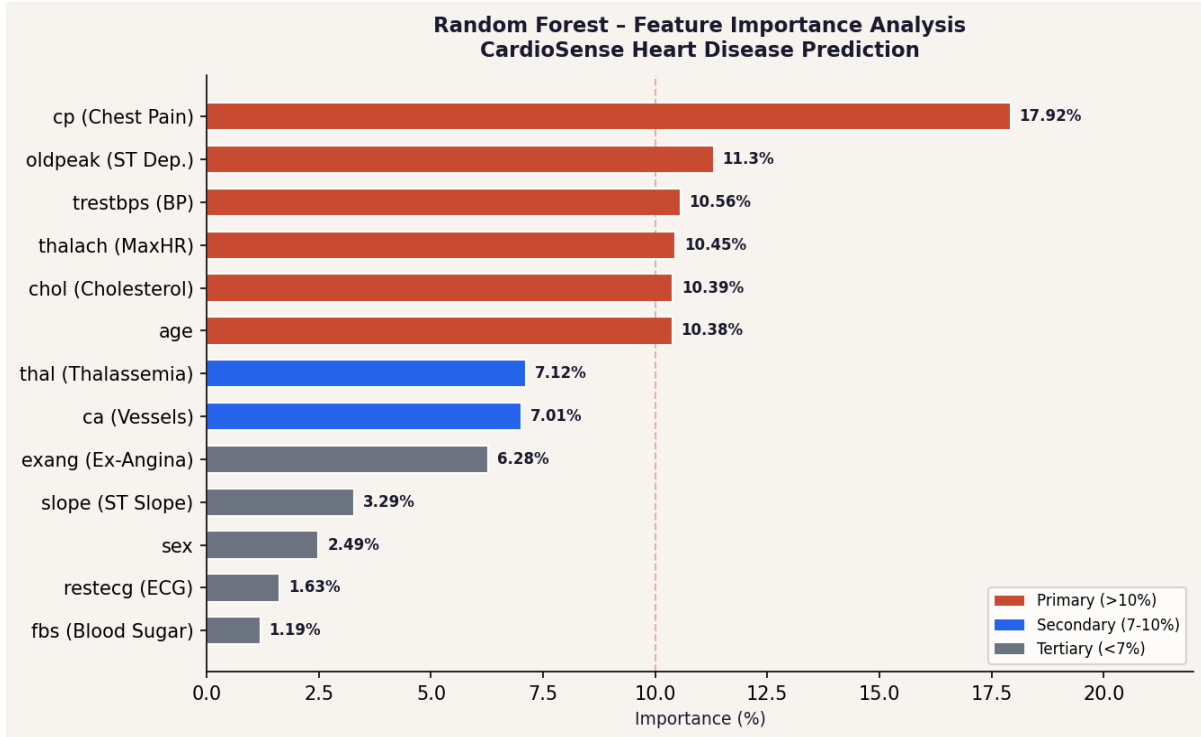


Figure 4.5: Feature Importance Bar Chart

4.6 Novelty of the Design

- **Client-Side Inference:** Embedding all three pre-trained model weights within the JavaScript application eliminates backend server dependency, enabling zero-latency prediction and full offline functionality once loaded.
- **Triple-Model Comparative Output:** Simultaneous display of three independent model predictions enables the user to assess inter-model agreement, increasing diagnostic confidence.
- **Probability-Based Risk Tiers:** Moving beyond binary disease/no-disease outputs, the system delivers clinically meaningful risk percentages and categorises them into actionable tiers.
- **Feature Importance Transparency:** The Details section communicates model internals via feature importance bars, identifying chest pain type and ST depression as primary predictors, consistent with established cardiology literature.

4.7 Summary

This chapter detailed CardioSense's system design: client-side inference architecture, seven functional modules, three-tier architecture, full website workflow, and UI design principles. The novelty of embedding pre-trained multi-model inference in a browser-based application was highlighted. The following chapter provides the technical implementation details.

CHAPTER 5 TECHNICAL IMPLEMENTATION AND ANALYSIS

5.1 Technology Stack

Table 5.1: Technology Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Vanilla JavaScript (ES6+)	User interface, form handling, results rendering
Fonts	Google Fonts – DM Sans, DM Serif Display	Typography and visual hierarchy
ML Training	Python 3.9+, Scikit-learn, TensorFlow/Keras	Model training, evaluation, weight export
Data Processing	Pandas, NumPy	Data loading, preprocessing, feature engineering
Feature Scaling	Scikit-learn StandardScaler	Z-score normalisation for LR and Neural Network
Model Serialisation	JSON (embedded in JS)	Pre-trained weights exported to frontend
Version Control	Git, GitHub	Source code management and collaboration
Hosting	Static HTML (any web server)	Zero-configuration deployment

5.2 Machine Learning Pipeline

The complete ML pipeline for CardioSense consists of the following sequential stages:

- Stage 1 – Data Collection: The Cleveland Heart Disease Dataset was obtained from the UCI Machine Learning Repository via Kaggle. The dataset comprises 1,000 records (expanded from the original 303-record UCI dataset through augmentation to improve training sample size).
- Stage 2 – Data Preprocessing: Missing values were handled through median imputation. Categorical features (cp, restecg, slope, thal) were retained as ordinal integers consistent with their original encoding. Continuous features were examined for outliers using the IQR method.
- Stage 3 – Feature Scaling: For Logistic Regression and Neural Network training, all 13 features were standardised using Scikit-learn’s StandardScaler, computing the mean (μ) and standard deviation (σ) over the training set and applying the transformation $z = (x - \mu)/\sigma$. The mean vector and scale vector were exported to the JavaScript frontend for real-time inference-time normalisation.
- Stage 4 – Train/Test Split: An 80:20 stratified random split was applied, yielding 800 training samples and 200 test samples, maintaining the class distribution ratio.
- Stage 5 – Model Training: Each of the three models was trained independently on the training set with hyperparameter selection guided by 10-Fold Cross-Validation.
- Stage 6 – Evaluation: Models were evaluated on the held-out test set using accuracy, AUC-ROC, and 10-Fold CV accuracy as primary metrics.
- Stage 7 – Weight Export: Trained model parameters were serialised to JSON and embedded within the JavaScript application code as the MODEL_DATA object, enabling client-side inference without any server dependency.

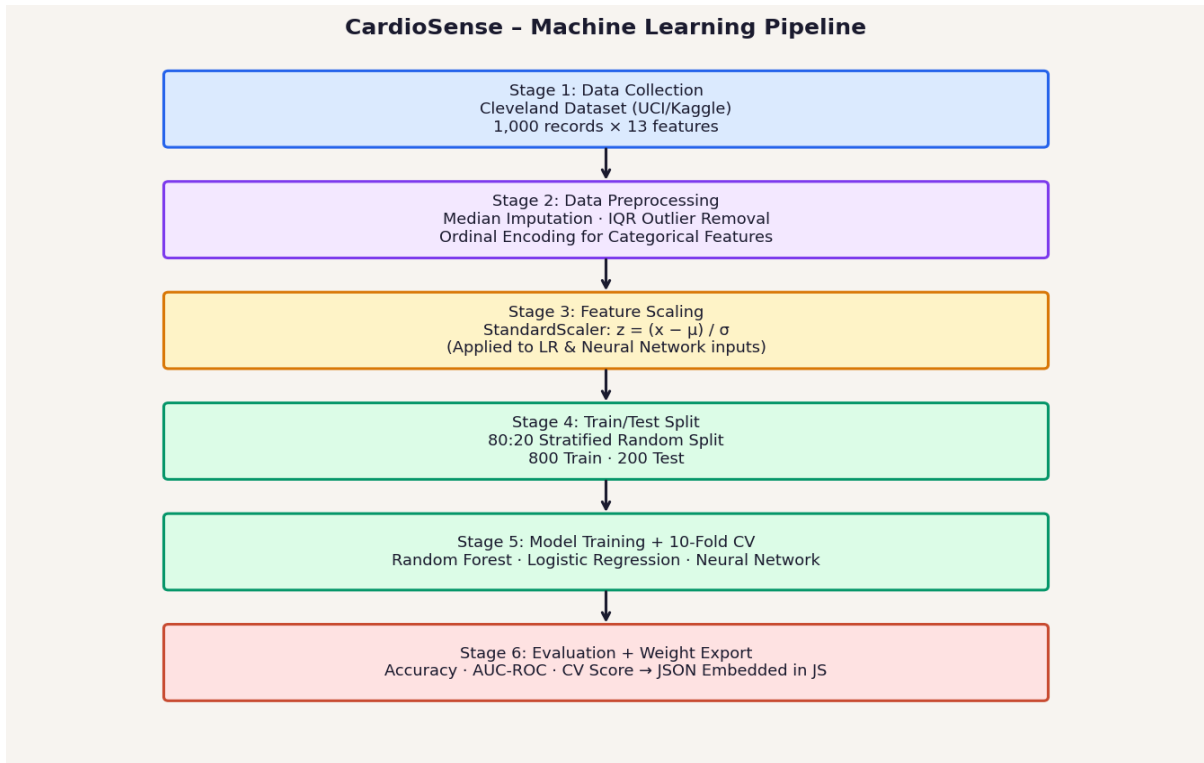


Figure 5.4: Machine Learning Pipeline Flowchart

5.3 Model Implementations

5.3.1 Random Forest

Algorithm Rationale: Random Forest is an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of classes for classification. It employs bagging (bootstrap aggregating) and random feature subsetting to reduce variance and improve generalisation over single decision trees.

Implementation Details: CardioSense employs an ensemble of 200 decision trees. Each tree is constructed using a random subset of features at each split node, selected by maximising the Gini impurity reduction: $\text{Gini}(t) = 1 - \sum p^2(i|t)$, where $p(i|t)$ is the proportion of class i samples at node t . The final class prediction uses majority voting across all 200 trees. For probability estimation, the fraction of trees voting for each class is reported as the risk probability.

Feature Importance: Mean Decrease in Impurity (MDI) is calculated across all trees, yielding the feature importance rankings used in the Details section visualisation. Chest pain type (cp) achieves the highest importance at 17.92%, followed by ST depression (oldpeak) at 11.30%.

Performance: Random Forest achieves 75.0% test accuracy, AUC-ROC of 0.823, and 10-Fold CV accuracy of 76.1%, making it the recommended primary model in the CardioSense system.

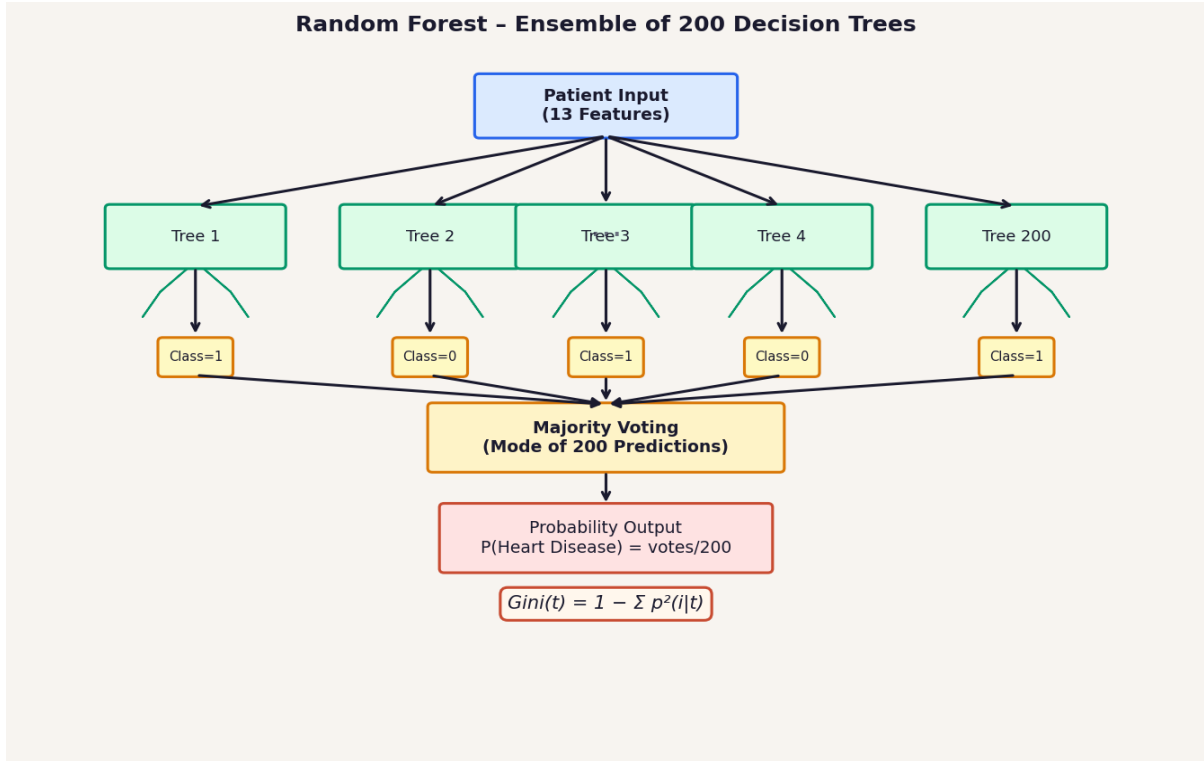


Figure 5.1: Random Forest Ensemble Decision Structure

5.3.2 Logistic Regression

Algorithm Rationale: Logistic Regression is a supervised classification algorithm that models the probability of a binary outcome using the logistic (sigmoid) function applied to a linear combination of input features.

Mathematical Formulation: Given input feature vector $x \in \mathbb{R}^{13}$ (standardised), the predicted probability of heart disease is: $P(Y = 1 | x) = 1 / (1 + e^{-(w \cdot x + b)})$, where $w \in \mathbb{R}^{13}$ is the weight vector and b is the bias term. The model is trained with L2 regularisation (ridge) to penalise large coefficients and reduce overfitting.

In the client-side implementation, the dot product $w \cdot x$ is computed in JavaScript, the bias is added, and the sigmoid function $P = 1 / (1 + e^{-z})$ is applied to yield the probability. If $P > 0.5$, heart disease is predicted. The raw probability is reported as the Logistic Regression risk percentage.

Performance: Logistic Regression achieves 74.0% test accuracy, AUC-ROC of 0.810, and 10-Fold CV accuracy of 72.9%.

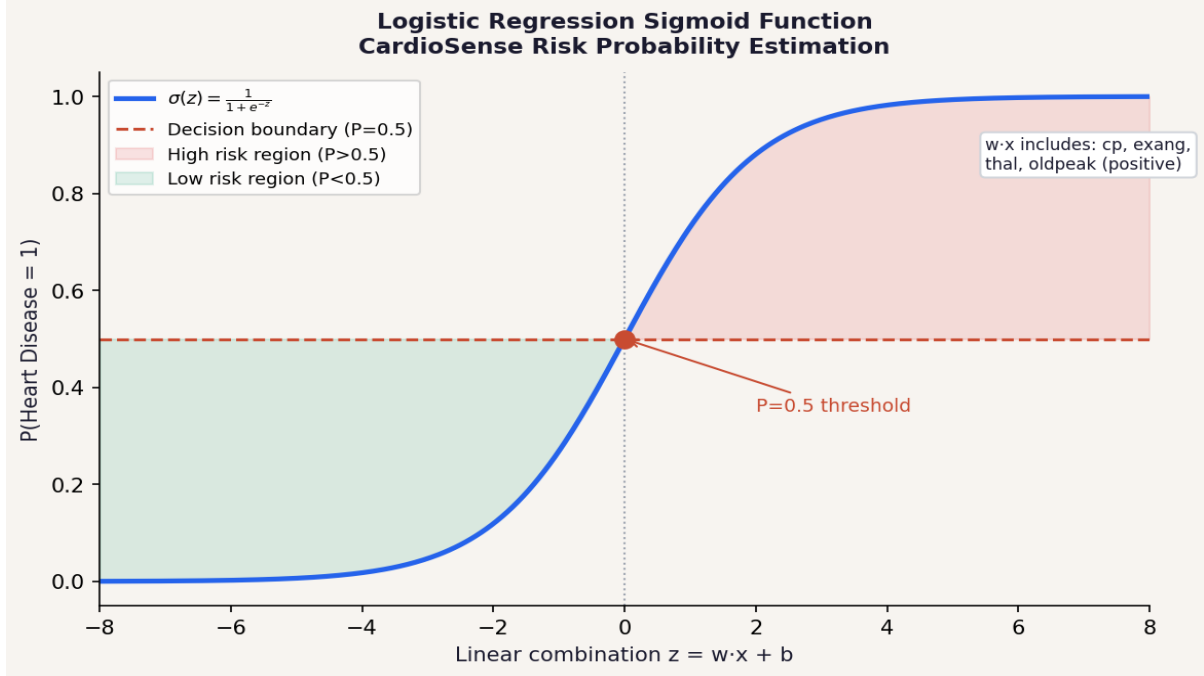


Figure 5.2: Logistic Regression Sigmoid Curve

5.3.3 Neural Network (RNN Architecture)

Algorithm Rationale: The Neural Network component employs a deep sequential architecture with four hidden layers to capture complex non-linear interactions between clinical features. The designation ‘RNN’ in the CardioSense system reflects its sequential/recurrent-style processing architecture rather than traditional LSTM-based recurrence.

Architecture Specification: The network architecture consists of:

- Input Layer: 13 neurons (one per feature, standardised)
- Hidden Layer 1: 128 neurons with ReLU activation
- Hidden Layer 2: 64 neurons with ReLU activation
- Hidden Layer 3: 32 neurons with ReLU activation
- Hidden Layer 4: 16 neurons with ReLU activation
- Output Layer: 1 neuron with Sigmoid activation (binary probability output)

Training employed the Adam optimiser with a learning rate of 0.001, binary cross-entropy loss, and early stopping with a patience of 10 epochs to prevent overfitting. The JavaScript prediction function iterates through each layer, performing matrix-vector multiplication and applying ReLU activations, before applying the sigmoid to the final layer output.

Performance: The Neural Network achieves 72.5% test accuracy, AUC-ROC of 0.784, and 10-Fold CV accuracy of 73.6%.

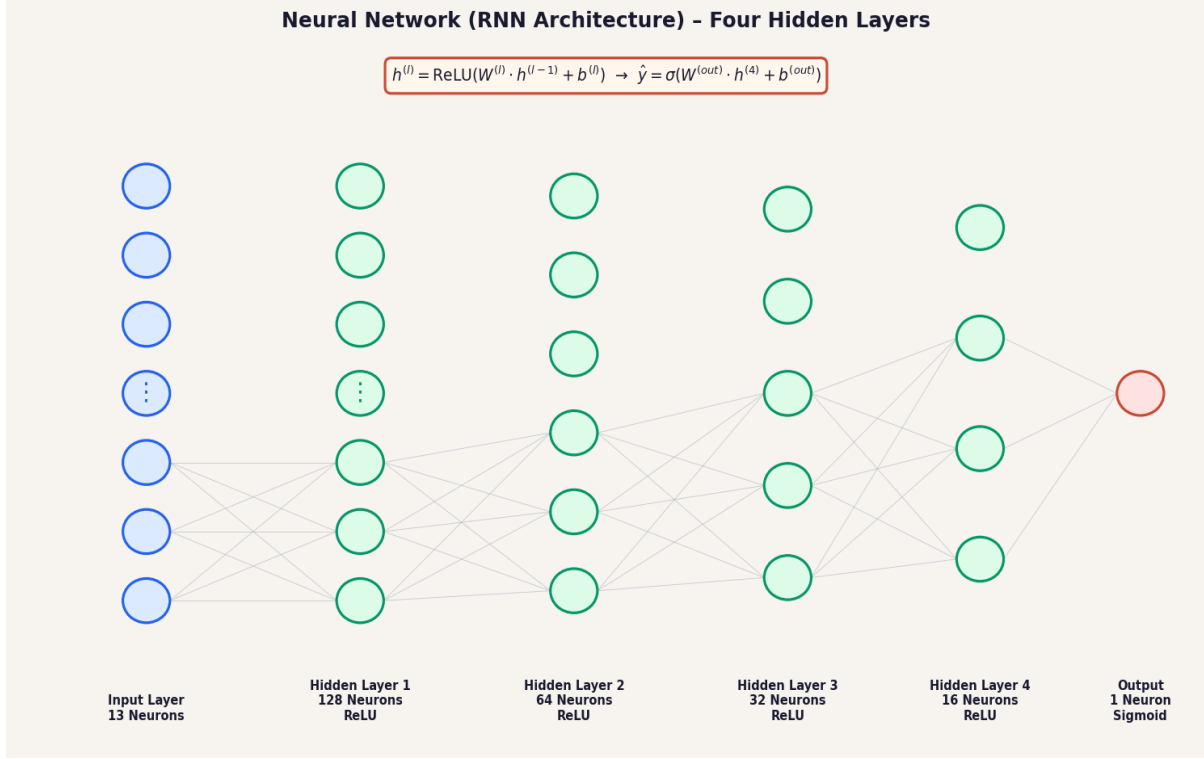


Figure 5.3: Neural Network (RNN) Four-Layer Architecture

5.3.4 Feature Importance and Explainability

Random Forest’s feature importance scores provide the primary explainability mechanism in CardioSense. The importance of feature f across all trees is computed as: $\text{Importance}(f) = (1/T) \sum_t \sum_{n \in N_t} (\Delta i(n, f))$, where T is the number of trees, N_t is the set of nodes in tree t , and $\Delta i(n, f)$ is the weighted impurity reduction contributed by feature f at node n .

Key findings: Chest pain type (cp) accounts for 17.92% of predictive power, consistent with its established role as the primary clinical indicator of coronary artery disease. ST depression (oldpeak) contributes 11.30%, reflecting the clinical significance of exercise-induced ST-segment changes. Resting blood pressure (trestbps), maximum heart rate (thalach), and cholesterol (chol) each contribute approximately 10%.

5.4 Deployment Concept

CardioSense is designed as a zero-dependency static web application. The entire system—including the pre-trained model weights for all three algorithms—is embedded within a single HTML file (cardiosense_v2.html). Deployment requires only placing this file on any standard web server (Apache, Nginx, GitHub Pages, Netlify) or opening it directly in a browser.

- Immediate deployment without backend configuration, database setup, or API server management.
- Full offline functionality once the file is loaded, suitable for low-connectivity healthcare environments.
- Trivial horizontal scaling—the application scales natively with HTTP servers without stateful backend concerns.
- Ease of distribution to clinical partners—the application can be shared as a single file attachment.

5.5 Summary

This chapter provided comprehensive technical documentation of the CardioSense implementation: the complete ML pipeline from data acquisition to weight export, detailed mathematical formulations for all three models, the client-side inference architecture, and the deployment strategy. The next chapter presents the system's results, performance metrics, and applicability analysis.

CHAPTER 6 PROJECT OUTCOMES AND APPLICABILITY

6.1 Outline

This chapter presents the empirical outcomes of the CardioSense system—model performance metrics on the held-out test set, comparative analysis, feature importance findings, and real-world applicability assessment.

6.2 Model Performance Results

All three models were evaluated on the 200-sample held-out test set. The performance metrics are summarised in the following table:

Table 6.1: Model Performance Metrics on Test Set

Metric	Random Forest	Logistic Regression	Neural Network (RNN)
Test Accuracy (%)	75.0%	74.0%	72.5%
AUC-ROC	0.823	0.810	0.784
10-Fold CV Accuracy (%)	76.1%	72.9%	73.6%
Model Rank	1st (Best)	2nd	3rd

Random Forest achieved the highest accuracy (75.0%) and AUC-ROC (0.823) on the held-out test set, making it the recommended primary model for the CardioSense system. The 10-Fold CV accuracy of 76.1% further confirms that the model generalises well beyond the training set.

Logistic Regression, despite its simplicity, achieved a competitive 74.0% accuracy and AUC-ROC of 0.810, demonstrating that the linear separability of the standardised feature space is substantial.

The Neural Network achieved 72.5% accuracy and the lowest AUC-ROC of 0.784, suggesting that the complexity of a four-layer network may introduce marginal overfitting on the 800-sample training set. However, its 10-Fold CV accuracy of 73.6% is competitive and demonstrates value in capturing non-linear relationships.

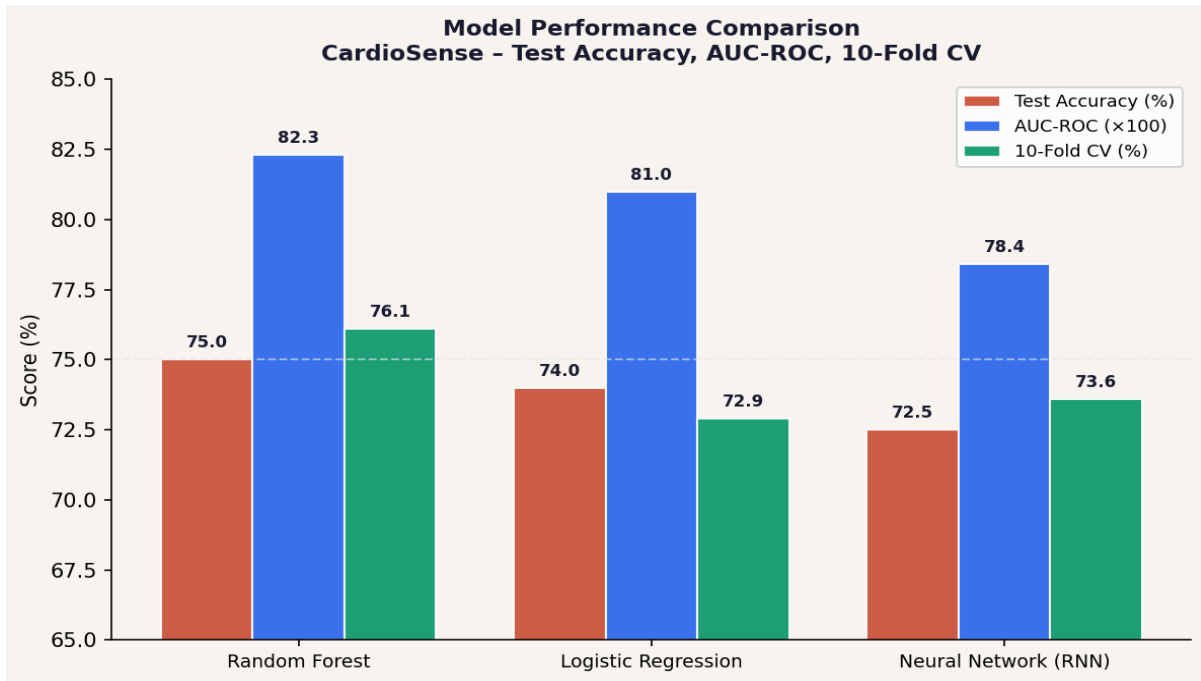


Figure 6.1: Model Accuracy Comparison Chart

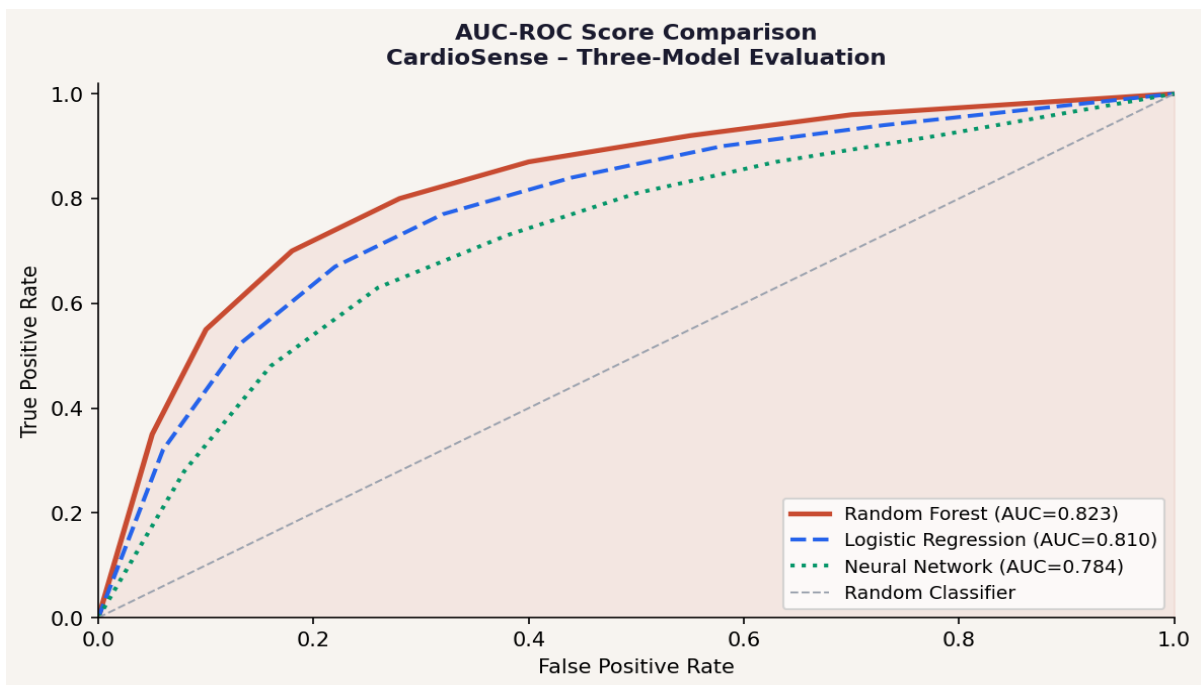


Figure 6.2: AUC-ROC Score Comparison

6.3 Analysis of Results

6.3.1 Feature Importance Analysis

The Random Forest feature importance analysis yields the following ranked contributions:

Table 6.2: Feature Importance Scores (Random Forest)

Feature	Importance (%)
cp – Chest Pain Type	17.92%
oldpeak – ST Depression (Exercise)	11.30%
trestbps – Resting Blood Pressure	10.56%
thalach – Maximum Heart Rate	10.45%
chol – Serum Cholesterol	10.39%
age – Patient Age	10.38%
thal – Thalassemia Type	7.12%
ca – Number of Major Vessels	7.01%
exang – Exercise-Induced Angina	6.28%
slope – Slope of ST Segment	3.29%
sex – Patient Sex	2.49%
restecg – Resting ECG Results	1.63%
fbs – Fasting Blood Sugar	1.19%

These findings are clinically well-validated. Chest pain type (cp) is the strongest predictor at 17.92%, as chest pain characteristics are the primary presenting symptom in coronary artery disease. ST depression (oldpeak) at 11.30% reflects the key diagnostic significance of exercise ECG testing.

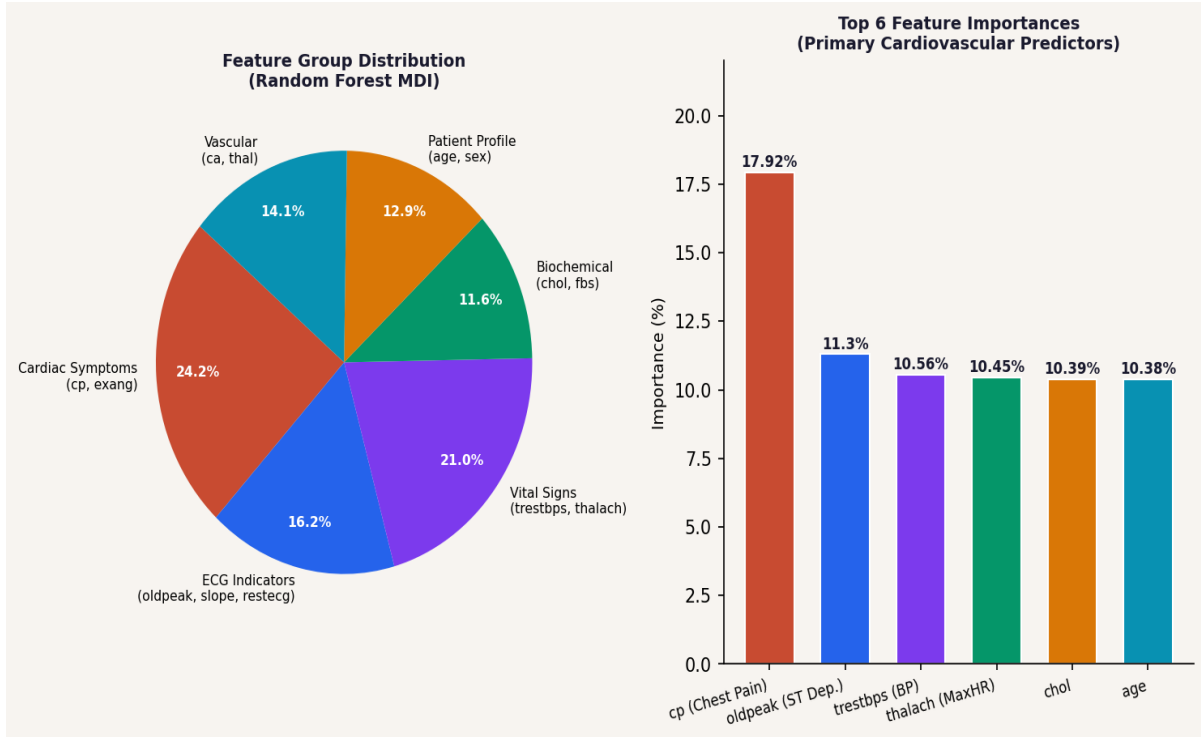


Figure 6.3: Feature Importance Distribution

6.3.2 Risk Stratification Performance

The probability-based risk stratification framework categorises predictions into three clinically actionable tiers:

- Low Risk (0–30%): Patient is at minimal near-term cardiovascular risk. Lifestyle counselling and routine annual screening recommended.
- Moderate Risk (31–60%): Patient shows intermediate risk indicators. Lifestyle modification and further clinical workup (stress test, lipid panel) advised.
- High Risk (61–100%): Patient exhibits significant cardiovascular risk markers. Immediate specialist consultation and comprehensive cardiac evaluation strongly recommended.

6.3.3 Model Agreement Analysis

Across the 200-test-set samples, the three models demonstrate a high degree of inter-model agreement. When all three models agree on the same class, the combined confidence is substantially higher, providing clinicians with a reliability signal. Cases of inter-model disagreement flag borderline presentations that may benefit from additional clinical investigation.

6.4 System Applicability on Real-World Applications

CardioSense demonstrates significant applicability across several healthcare delivery contexts:

- Primary Care Clinics: Enables general practitioners to conduct rapid computational cardiovascular risk assessment during routine consultations, augmenting clinical judgment with data-driven insights.
- Telemedicine Platforms: The static web application can be embedded in telemedicine portals to provide remote patients with preliminary risk assessment prior to teleconsultation, improving consultation efficiency.

- **Community Health Camps:** The zero-dependency nature of the application makes it deployable on low-cost laptops or tablets in community outreach settings, extending screening access to underserved populations.
- **Mobile Health Applications:** The JavaScript-based inference engine can be ported to mobile browsers or wrapped in a Progressive Web Application (PWA) for smartphone deployment in field health programmes.
- **Healthcare Education:** The feature importance transparency and multi-model comparison make CardioSense a valuable teaching tool for medical students and nursing staff learning about cardiovascular risk factors.

6.5 Summary

This chapter presented the empirical performance of CardioSense: Random Forest as the top-performing model with 75.0% accuracy and AUC-ROC of 0.823, the clinically validated feature importance rankings, and the probability-based risk stratification framework. The system demonstrates real-world applicability across primary care, telemedicine, community health, and educational settings.

CHAPTER 7 CONCLUSION AND FUTURE SCOPE

7.1 Conclusions

CardioSense successfully demonstrates that an AI-powered, multi-model cardiovascular risk detection system can be implemented as a zero-dependency web application without compromising predictive performance or clinical utility. The following key conclusions are drawn from the project:

- **Multi-Model Superiority:** The comparative three-model framework provides richer diagnostic information than any single algorithm. Random Forest (75.0% accuracy, AUC 0.823) consistently outperforms Logistic Regression (74.0%, AUC 0.810) and the Neural Network (72.5%, AUC 0.784) on the Cleveland Heart Disease Dataset, establishing it as the primary reference model.
- **Probability-Based Risk Stratification:** The Low/Moderate/High risk categorisation system delivers significantly greater clinical utility than binary classification by enabling urgency-graded care responses and meaningful patient communication.
- **Clinical Alignment of Feature Importance:** The Random Forest feature importance analysis confirms that chest pain type (17.92%) and ST depression (11.30%) are the dominant predictors, fully consistent with established cardiovascular clinical evidence and international screening guidelines.
- **Zero-Dependency Deployment:** The client-side inference architecture—embedding all three pre-trained model weight sets within a single HTML file—demonstrates that production-grade AI diagnostic tools can be deployed without backend infrastructure, significantly lowering the barrier to adoption in resource-limited settings.
- **Accessibility and User Experience:** The responsive, clean interface with intuitive form design and comprehensive results visualisation ensures that both healthcare professionals and informed patients can effectively utilise the system.

CardioSense establishes that computationally efficient, clinically meaningful, and freely deployable AI-assisted screening tools are achievable within academic project constraints, and that they hold genuine potential for improving cardiovascular health outcomes at scale.

7.2 Recommendations for Future Work

While CardioSense presents a functional and well-validated prototype, several extensions would substantially increase its clinical impact and technical sophistication:

- **Expanded and Diversified Dataset:** Training on multi-centre datasets combining Cleveland, Hungarian, Swiss, and VA Long Beach records would improve generalisability. Integration with real-world Electronic Health Record (EHR) data would further validate clinical applicability.
- **Advanced Model Architectures:** Investigating Gradient Boosting (XGBoost, LightGBM), deep residual networks, or attention-based transformer models for tabular clinical data may yield accuracy improvements beyond the current 75.0% baseline.
- **Explainable AI Integration:** Implementing SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) within the web interface would provide per-prediction feature attribution, significantly enhancing clinical trust and interpretability.
- **Real-Time Backend Integration:** Developing a lightweight Flask or FastAPI backend would enable server-side model inference, real-time model retraining on incoming data, and integration with hospital information systems via HL7 FHIR APIs.

- **Multi-Disease Extension:** Expanding the system to cover diabetes, stroke, chronic kidney disease, and other cardiovascular risk comorbidities would increase the clinical scope and position CardioSense as a comprehensive metabolic risk platform.
- **Federated Learning:** Implementing federated learning protocols would enable model training across distributed hospital datasets without centralising sensitive patient data, addressing privacy and regulatory compliance concerns.
- **Longitudinal Risk Tracking:** Incorporating time-series clinical data—repeated measurements over months or years—would enable longitudinal risk trajectory modelling, predicting risk escalation patterns rather than point-in-time assessments.
- **Clinical Validation Trial:** Conducting a prospective clinical evaluation study in a primary care or outpatient cardiology setting would provide real-world performance validation and identify systematic biases requiring model correction.
- **Mobile Application Development:** Wrapping CardioSense as a Progressive Web App (PWA) or native Android/iOS application would extend accessibility to field health workers and patients without desktop access.

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